3.5.6 Wrap-Up: Ticket Out the Door

1. Rewrite each expression as an equivalent exponential expression with positive exponents.

   a. $\left(\frac{-13}{2}\right)^{-3} =$ ___________

   b. $\left(\frac{1}{5}\right)^{-8} =$ ___________

2. Generate an equivalent expression using the laws of exponents. Express your answer with positive exponents.

$$\left(\frac{-5}{4}\right)^{-7} \cdot \left(\frac{-5}{4}\right)^{-3}$$

Unit 3, Lesson 6: Evaluating Numerical Expressions with Integer Exponents

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- [ ] I can evaluate a numerical expression using the laws of exponents.

3.6.1 Warm-Up: Negative Exponents

1. Place a digit in each blue box to make the statement true. Use only digits 0 to 9, at most one time each.

$$\square^{-\square} = \frac{\square}{\square\,\square}$$

3.6.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions

1. Melanie and Adonis each evaluated the expression $\frac{(5)^{-2}(5)^{5}}{5}$. Each student's steps are shown.

Melanie	Adonis
$\frac{(5)^{-2}(5)^{5}}{5}$	$\frac{(5)^{-2}(5)^{5}}{5}$
$= \frac{5^{3}}{5}$	$= \frac{\frac{1}{5^{2}} \cdot (5)^{5}}{5}$
$= 5^{2}$	$= \frac{\frac{1}{25} \cdot 3125}{5}$
$= 25$	$= \frac{125}{5}$
	$= 25$

Discuss with your partner how Melanie's steps and Adonis' steps are similar and different from each other. Then, summarize your discussion in the table.

Similarities	Differences

When evaluating exponential expressions, it is helpful to apply the laws of exponents before evaluating.

2. Complete the steps to evaluate the expression $\frac{(6^2)^{-3}}{6^{-5}}$ by completing the description of each step and writing the resulting equivalent expression on the line that follows.

   a. Apply ○ product of power law. ○ power of a power law. __________

   b. Apply ○ quotient of power law. ○ power of a power law. __________

   c. Evaluate. __________

3. Consider the expression $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

   a. Discuss with your partner which laws of exponents could be used to evaluate the expression.

   b. Evaluate $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

3.6.3 Try It: Evaluating Expressions

1. Evaluate each expression. Show your work in the space provided. Then, check off the exponent laws applied to each expression from the list provided.

   a. $\left(\frac{3}{4}\right)^{-2} \cdot 3$

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

   b. $\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$

Exponent Laws Applied
☐ identity exponent
☐ negative exponent
☐ power of a power
☐ power of a product
☐ power of a quotient
☐ product of powers
☐ quotient of powers
☐ zero exponent

3.6.4 Guided Practice: Evaluating Exponential Expressions

1. Danny and Reina evaluated the expression $\dfrac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$.

   They found the same value but used different processes. Their work is shown.

Danny's Work	Reina's Work
$\dfrac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$	$\dfrac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$
$= \dfrac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$	$= \dfrac{\left(\frac{3}{4}\right)^{-3} \cdot 1}{\left(\frac{3}{4}\right)^{-2}}$
$= \dfrac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$	$= \dfrac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$
$= \left(\frac{3}{4}\right)^{-3-(-2)}$	$= \left(\frac{3}{4}\right)^{-3-(-2)}$
$= \left(\frac{3}{4}\right)^{-1}$	$= \left(\frac{3}{4}\right)^{-1}$
$= \frac{4}{3}$	$= \frac{4}{3}$

   a. With your partner, discuss and compare how each student evaluated the expression.

   b. Use words or phrases from the bank to complete the descriptions of each student's work. Some words or phrases will be used more than once and others not at all.

zero	identity
power of a power	negative
power of a product	product of powers
power of a quotient	quotient of powers

   Danny applied the ______________________________ exponent law first by rewriting $\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^{0}$ as $\left(\frac{3}{4}\right)^{-3}$. Then, he applied the ____________________________ exponent law, subtracting the exponents -3 and -2. Finally, he applied the _______________________ exponent law to find the value of the expression.

   Reina applied the ______________________________ exponent law first by re-writing $\left(\frac{3}{4}\right)^{0}$ as 1. Then, she multiplied $\left(\frac{3}{4}\right)^{-3}$ by 1. Next, she applied the ______________________________ exponent law by subtracting the exponents -3 and -2. Finally, she applied the _______________________ exponent law to find the value of the expression.

While the students applied the laws differently, they both generated equivalent expressions in each step to find the value of the expression.

3.6.5 Collaboration: Choose Your Own Adventure!

1. Consider the expression $\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$.

   a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

   - ☐ zero exponent
   - ☐ identity exponent
   - ☐ negative exponent
   - ☐ power of a power
   - ☐ product of powers
   - ☐ power of a product
   - ☐ quotient of powers
   - ☐ power of a quotient

   b. Evaluate the expression beginning with a different first step than your partner.

   c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

2. Consider the expression $\frac{\left(\frac{3}{2}\right)^{-5} \cdot \left(\left(\frac{2}{3}\right)^7\right)^0}{\left(\left(\frac{2}{3}\right)^2\right)^3}$.

   a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

   - ☐ zero exponent
   - ☐ identity exponent
   - ☐ negative exponent
   - ☐ power of a power
   - ☐ product of powers
   - ☐ power of a product
   - ☐ quotient of powers
   - ☐ power of a quotient

   b. Evaluate the expression beginning with a different first step than your partner.

   c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

3.6.6 Wrap-Up: Error Analysis

1. Two students each evaluated an expression. Their work is shown in the table. Each student made a mistake. Circle the step where each student made an error and describe the mistake they made.

Problem	Describe the Mistake
$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} \cdot 5^{-1}$ $= \left(\frac{1}{5}\right)^{-8} \cdot 5^{-1}$ $= 5^8 \cdot 5^{-1}$ $= 5^7$ $= 78{,}125$	
$\left(\frac{3}{2}\right)^{-3} \cdot (2)^3$ $= \frac{2^{-3}}{3^{-3}} \cdot 2^3$ $= \frac{2^0}{3^{-3}}$ $= \frac{1}{3^{-3}}$ $= 3^3$ $= 27$	

Unit 3, Lesson 7: Equivalent Expressions

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ I can determine equivalency using the laws of exponents.